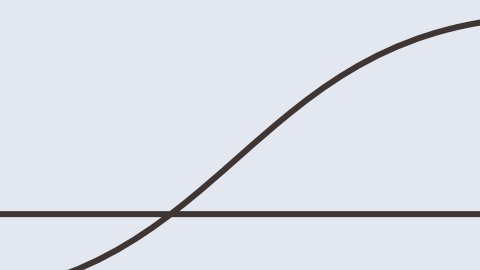


Compost Irrigation System

Compost Critters: Olivia, Ben, Chase, Paul, & Maraki



Clients



Sustainable Claremont

Working for Environmental
Economic and Social Sustainability

Background/Design need

- Multiple compost piles of different shapes/sizes
- Compost needs water for microorganisms
- Current solution: hand watering
 - Manual labor
 - Takes 3 hours



Sustainable Claremont requires a reliable and low-labor method for watering its community compost piles. The solution needs to effectively distribute water throughout piles of varying sizes up to 4x4x4 ft, without flushing out the microorganisms needed for decomposition.

Objectives

- 1.) Maximize water distribution
- 2.) Minimize set-up time
- 3.) Maximize transportability
- 4.) Minimize deformation of compost pile

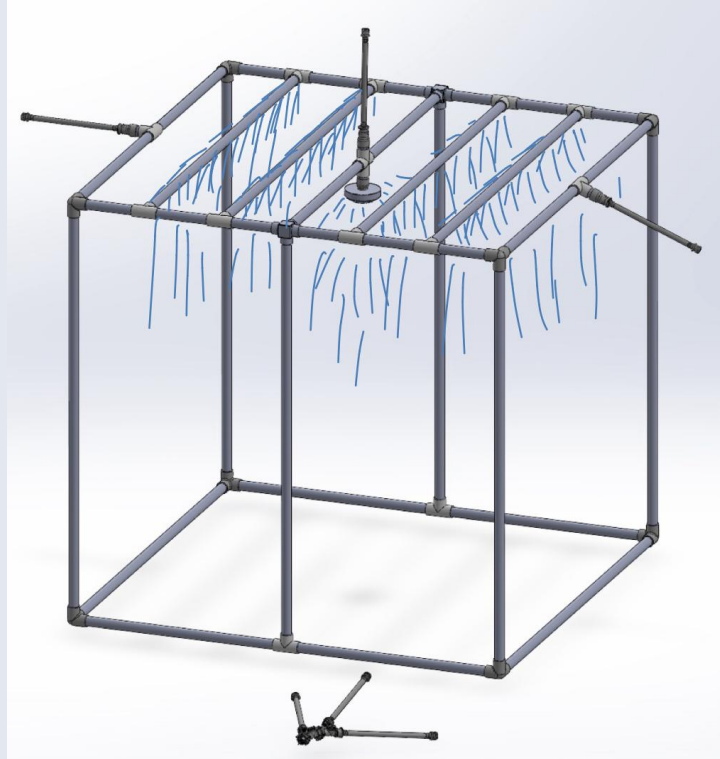
Constraints

- 1.) Must operate autonomously once turned on
- 2.) Must be safe for children to be around
- 3.) Must not use water pressure higher than 80 PSI to preserve microorganisms
- 4.) Must be compatible with every range of pile up to 4x4x4ft compost pile sizes
- 5.) Must withstand rain, sun, and wind for 24 hours a day
- 6.) Must be compatible with existing water sources
- 7.) Must cost less than \$150 to design
- 8.) Must withstand water pressure without breaking

Prototype 1



Pivot



Head Prototypes



Head material



Thread size to connect
to frame



Size and hole distribution

Water Distribution test



Top left Green Cup Average Water Height: 1.94 cm

Blue Cup Average Water Height: 1.99 cm

Pink Cup Average Water Height: 2.01 cm

Orange Cup Average Water Height: 1.46 cm

Middle Green Cup Average Height: 6.04 cm

- Top Left Green, Blue, and Pink Cups had a difference of less than 10% in water height.
- Middle Green was 3x as much as outer which was target
- Orange Cups differed by less than 30%
- Leak in head was discovered on Orange cup side- Resin Epoxy fixed issue

Final Head Design



Weight Test



Structure Test Data

Weight (lbs)	Displacement (cm)
5	0.5
10	1.125
15	1.75
20	2.5
25	3.1
30	3.875
35	4.75
40	5.5

Before Structure Update

Weight (lbs)	Displacement (cm)
5	0
10	0
15	0
20	0
25	0
30	0
35	0
40	0

After Structure Update

Final prototype



Key Finding

- Client tested our prototype and was satisfied with the internal distribution of water compost pile.

This result validates our design choices throughout this process because it shows that we were able to achieve our main objective of maximizing water distribution. During this test our client told us that having a lower water pressure allows for the water be distributed more evenly throughout the pile instead of just sitting on the top.

Future recommendations

- Adjusts to uneven ground.
- Collapsible without deconstruction.
- Allows for greater variation in water pressure.